

Inheritance Questions

1. Bank Management System

Create a Bank class with:

- balance variable
- deposit()
- withdraw()
- check_balance()

Create a User class that inherits Bank and displays the user's name. Perform deposit, withdrawal, and balance check.

2. Employee Salary System

Create an Employee class with:

- emp_name
- salary
- display_details()

Create a Manager class that inherits Employee and adds a bonus(). Display the total salary.

1. Student Result System

Create a Student class with:

- Name
- marks
- display_marks()

Create a Result class that inherits Student and calculates whether the student has passed or failed.

4. Food Ordering System Using Multilevel Inheritance

Class 1: Restaurant

- Create a method menu(item) that returns the price of the selected food item.

Class 2: FoodCourt (*inherits Restaurant*)

Create the following methods:

- display_menu() – Display the available food items.
- order() – Accept the food item from the user and allow multiple orders.
- billing() – Display the total bill and add a packing charge of ₹20.

Class 3: Customer (*inherits FoodCourt*)

- Create an object of the Customer class.
- Call the order() method.

5. Movie Ticket Booking System Using Multilevel Inheritance

Class 1: Movie

- Create a method ticket(movie) that returns the ticket price.

Class 2: Booking (*inherits Movie*)

Create the following methods:

- movies() – Display the available movies.
- selection() – Allow the user to book multiple tickets.
- billing() – Display the total amount and add a booking charge of ₹30.

Class 3: Customer (*inherits Booking*)

- Create an object and call the selection() method.

6. Online Course Enrollment System Using Multilevel Inheritance

Class 1: Course

- Create a method `fee(course)` that returns the course fee.

Class 2: Academy (*inherits Course*)

Create the following methods:

- `courses()` – Display available courses.
- `enroll()` – Allow the user to enroll in multiple courses.
- `billing()` – Display the total fee and add a registration fee of ₹100.

Class 3: Student (*inherits Academy*)

- Create an object and call the `enroll()` method.

7. Cab Booking System Using Hierarchical Inheritance

Class 1: Cab

- Create methods to calculate the fare for **Bike**, **Auto**, and **Car** rides.

Class 2: Uber (*inherits Cab*)

- Create the methods `menu()`, `booking()`, and `billing()`.
- Add **10% GST** and apply a **15% discount** if the bill is above ₹1000.

Class 3: Ola (*inherits Cab*)

- Create the methods `menu()`, `booking()`, and `billing()`.
- Add **12% GST** and apply a **20% discount** if the bill is above ₹1500.

Driver Code

- Ask the user to choose **Uber** or **Ola** and call the `booking()` method.

8. Grocery Shopping System Using Hierarchical Inheritance

Class 1: Grocery

- Create methods to return the price of **Rice**, **Sugar**, and **Oil**.

Class 2: Dmart (*inherits Grocery*)

- Create the methods `items()`, `shopping()`, and `billing()`.
- Add **5% GST** and apply a **10% discount** if the bill is above **₹2000**.

Class 3: Reliance Smart (*inherits Grocery*)

- Create the methods `items()`, `shopping()`, and `billing()`.
- Add **5% GST** and apply a **15% discount** if the bill is above **₹2500**.

Driver Code

- Ask the user to choose the supermarket and call the `shopping()` method.

9. Bus Ticket Booking System Using Hierarchical Inheritance

Class 1: Bus

- Create methods to return the fare for **Sleeper**, **Semi-Sleeper**, and **AC** buses.

Class 2: RedBus (*inherits Bus*)

- Create the methods `routes()`, `booking()`, and `billing()`.
- Add **10% GST** and a reservation charge of **₹30**.

Class 3: AbhiBus (*inherits Bus*)

- Create the methods `routes()`, `booking()`, and `billing()`.
- Add **10% GST** and a reservation charge of **₹20**.

Driver Code

- Ask the user to choose the platform and call the `booking()` method.

10. ATM System Using Multiple Inheritance

Class 1: SBI

- Create the methods deposit(amount) and check_balance().

Class 2: UnionBank

- Create the methods withdraw(amount) and mini_statement().

Class 3: ATM (*inherits SBI and UnionBank*)

- Create the methods menu() and transaction().
- Allow the user to perform banking operations.

Driver Code

- Create an object of the ATM class.
- Call the transaction() method.

11. Paytm Application Using Multiple Inheritance

Write a Python program to implement a **Paytm Application** using **multiple inheritance**.

Class 1: MobileRecharge

- Create the methods recharge_plans() and mobile_recharge().

Class 2: BusTicketBooking

- Create the methods display_buses() and book_ticket().

Class 3: ElectricityBills

- Create the methods bill_details() and pay_bill().

Class 4: Paytm (*inherits MobileRecharge, BusTicketBooking, and ElectricityBills*)

Create the following methods:

- menu() – Display the available services.
- services() – Allow the user to choose and use any service (Mobile Recharge, Bus Ticket Booking, or Electricity Bill Payment).